

Biology, 2018 EC, entrance exam, version 4

Grade 12 · Biology · 2018 E.C.

Time allowed: 120 minutes

Questions: 59

Total marks: 59

Name _____ Grade _____ ID No. _____ Date _____

1. What is the primary function of the Archaea (methanogens) in a biogas fermenter? (1)
 - A. Convert organic waste to alcohol
 - B. Produce methane gas
 - C. Break down raw organic waste into acids
 - D. Filter the liquid feedstock
2. Which of the following is considered as non-renewable natural resource? (1)
 - A. Petroleum
 - B. Geothermal energy
 - C. Biomass
 - D. Timber
3. What is the primary goal of bio-fortification in agricultural biology? (1)
 - A. To increase the shelf life of harvested vegetables.
 - B. To make plants resistant to high-temperature stress.
 - C. To produce bio-pesticides using natural soil bacteria.
 - D. To reduce hidden hunger by increasing micronutrient levels in staple crops
4. A scientist is trying to remove a fatty stain from a delicate silk garment. Which enzyme should be included in the detergent? (1)
 - A. Protease
 - B. Pectinase
 - C. Lipase
 - D. Amylase
5. If a dairy producer wants to make yogurt, which combination of bacteria is required? (1)
 - A. Streptococcus thermophilus and Lactobacillus bulgaricus
 - B. Streptococcus cremoris and Lactobacillus acidophilus
 - C. Propionic bacteria and Saccharomyces
 - D. Pediococcus cerevisiae and Penicillium
6. What is the main role of Saccharomyces cerevisiae in bread making? (1)
 - A. It kills pathogenic bacteria that cause bread mold
 - B. It respires aerobically to produce CO₂, which leaves the dough
 - C. It adds protein to the flour, which raises the dough
 - D. It prevents the dough from becoming elastic
7. Which statement best describes Bio-mining (Biohydrometallurgy)? (1)
 - A. Using heavy machinery to extract ores
 - B. Analyzing soil for chemical pollution
 - C. Turning food waste into compost
 - D. Using microbes to leach metals from ores
8. What is the relationship between Biochemical Oxygen Demand (BOD) and water pollution? (1)
 - A. Low BOD indicates high organic content and low oxygen, shows high pollution
 - B. High BOD indicates the water is safe for drinking
 - C. High BOD indicates high organic content and low oxygen, signaling pollution
 - D. BOD has no relationship with pollution levels

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9. How does distillation affect the alcohol content in spirits compared to beer? (1)
- A. It lowers its alcohol content to 5% B. It keeps it the same as wine
C. It raises its alcohol content to 40%-50% D. It removes all ethanol from the drink
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10. Why is Bio-transformation considered a better alternative to traditional chemical waste disposal? (1)
- A. It is more expensive but faster to degrade pollutant substances
B. It uses organisms to degrade toxic pollutants into harmless substances.
C. It creates more waste that can be recycled
D. It only works on non-biodegradable materials
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11. Imagine you are a conservation biologist tasked with restoring a degraded local ecosystem. Which combination of strategies would best ensure long-term sustainability? (1)
- A. Introducing exotic species to increase variety and using chemical fertilizers.
B. Establishing a gene bank for local seeds and implementing habitat restoration programs
C. Clearing all old vegetation and replacing it with fast-growing genetically modified timber.
D. Building a fence around the area and preventing any human interaction or education.
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12. Which bacterial structure is used primarily for attachment to surfaces? (1)
- A. Flagella B. Capsule
C. Pili (fimbriae) D. Plasmid
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13. Which domain consists of organisms that often live in extreme environments like volcanic vents? (1)
- A. Eukarya B. Bacteria
C. Archaea D. Protozoa
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14. What type of symmetry does the Tobacco Mosaic Virus (TMV) have? (1)
- A. Helical B. Icosahedral
C. Complex D. Spherical
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15. This item is based on the following choices of feeding habit employed by organisms. I. Synthesizing their own nutrients through the process of photosynthesis. II. Absorbing nutrient dissolved in the surrounding environment. III. Obtaining food living on or in other organisms like plants or animals. Which method/s do fungi use to get their nutrition? They employed (1)
- A. I and III only. B. I and II only.
C. II and III only. D. I, II and III
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16. Why are viruses considered to be on the border between living and non-living? (1)
- A. They have a cell wall but no nucleus.
B. They are crystallizable and inert outside a host cell.
C. They can only reproduce asexually.
D. They consist of both DNA and RNA.
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17. How do Gram-positive bacteria differ from Gram-negative bacteria regarding their cell wall? (1)
- A. Gram-positive have a thin peptidoglycan layer and an outer membrane.
B. Gram-negative have a thick peptidoglycan layer and an outer membrane.
C. Gram-positive have a thick peptidoglycan layer and lack an outer membrane.
D. Gram-negative have a thin peptidoglycan layer and lack an outer membrane.

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- 18.** What is the primary role of normal microbiota in protecting the host? (1)
- A. They produce antibiotics that kill the host's cells.
 - B. They help in the replication of viruses.
 - C. They increase the host's body temperature.
 - D. They compete with pathogens for space and nutrients.
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- 19.** What is the defining characteristic of Obligate anaerobes? (1)
- A. They require high levels of oxygen to survive.
 - B. They are killed by the presence of oxygen.
 - C. They can grow with or without the presence of oxygen.
 - D. They only grow in extreme acidic conditions.
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- 20.** What is the function of reverse transcriptase enzyme in retrovirus? It converts: (1)
- | | |
|-----------------------|-----------------------|
| A. RNA to DNA | B. DNA to RNA |
| C. DNA to another DNA | D. DNA to nucleotides |
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- 21.** A thermophilic Archaeon like *Thermus aquaticus* is commercially valuable because it is a source of: (1)
- A. Antibiotics resistant to high temperatures.
 - B. Enzymes stable at high temperatures for use in PCR.
 - C. Detergents that function in acidic conditions.
 - D. Methane gas for alternative energy.
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- 22.** If you apply 5% bleach to a table to kill microbes, you are performing: (1)
- | | |
|-----------------|------------------|
| A. Disinfection | B. Sanitization |
| C. Antiseptics | D. Sterilization |
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- 23.** A patient has a 'bloody cough' and 'chest pain' after exposure to moldy hay. What is the likely fungal infection? (1)
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|------------------|----------------|
| A. Thrush | B. Candidiasis |
| C. Aspergillosis | D. Ringworm |
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- 24.** Bacterial conjugation is a process that involves the transfer of genetic material between two cells and is significant because it: (1)
- A. Is the primary method of bacterial reproduction.
 - B. Leads to the formation of endospores.
 - C. Increases genetic diversity within a population.
 - D. Always results in the death of the donor cell
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- 25.** A scientist injects a healthy lab mouse with a pure culture of a suspected pathogen and the mouse subsequently develops a disease. Which Koch's Postulate is directly demonstrated by this step? (1)
- | | |
|------------------|-----------------|
| A. Postulate I | B. Postulate II |
| C. Postulate III | D. Postulate IV |
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- 26.** Formulate an argument for why 'Sanitization' is different from 'Sterilization.' (1)
- A. They are exactly the same.
 - B. Sterilization is only for food, Sanitization used only for tools.
 - C. Sanitization kills all life; sterilization reduces microbes to safe levels.
 - D. Sanitization reduces microbes to safe levels; sterilization kills all microbes

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- 27.** Predict what would happen to the global carbon cycle if all decomposer fungi and bacteria were removed from ecosystems. (1)
- A. Carbon dioxide levels in atmosphere would increase.
 - B. Carbon dioxide levels in atmosphere would decrease
 - C. Plants would grow faster due to high CO₂ in atmosphere
 - D. Nothing would change on CO₂ level in atmosphere.
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- 28.** Which of the following is NOT a product of the Krebs cycle per turn? (1)
- A. ATP (or GTP)
 - B. CO₂
 - C. Acetyl CoA
 - D. FADH₂
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- 29.** In the equation for aerobic respiration, $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$, the glucose molecule is: (1)
- A. Reduced.
 - B. Oxidized.
 - C. Phosphorylated.
 - D. Hydrolyzed.
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- 30.** A coupled reaction in metabolism involves: (1)
- A. Two anabolic reactions occurring simultaneously.
 - B. Two catabolic reactions occurring simultaneously.
 - C. An exergonic reaction driving an endergonic reaction.
 - D. An endergonic reaction driving an exergonic reaction.
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- 31.** Select the incorrect statement about glycolysis. (1)
- A. Glycolysis occurs in the mitochondria
 - B. Glycolysis is the first step in both aerobic and anaerobic respiration
 - C. It ends up with production of 2 ATP(net), 2 NADH, and 2 pyruvate
 - D. It can occur independently of the presence of oxygen.
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- 32.** Which of the following metabolic processes is an example of a catabolic pathway? (1)
- A. The conversion of glucose to pyruvate during glycolysis.
 - B. The formation of starch from glucose molecules.
 - C. The synthesis of proteins from amino acids
 - D. The fixation of CO₂ into organic molecules
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- 33.** In the context of the Krebs cycle, why is it also called the 'Citric Acid Cycle'? (1)
- A. Citric acid is the enzyme that starts the reaction.
 - B. Citric acid is the first stable 6-carbon compound formed.
 - C. The cycle only occurs in citrus fruits.
 - D. Citric acid is the waste product released at the end
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- 34.** What is the purpose of the Link Reaction? (1)
- A. To produce 4 ATP molecules.
 - B. To split glucose into two 3-carbon sugars by generating ATP.
 - C. To regenerate NAD⁺ for continuity of glycolysis.
 - D. To convert pyruvate into Acetyl-CoA for entry into the Krebs cycle.

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- 35.** Which of the following is the main difference between cyclic and noncyclic photophosphorylation? (1)
- A. They use different electron acceptors
 - B. Noncyclic photophosphorylation occurs in the stroma
 - C. Only photosystem I is used during the cyclic process
 - D. Noncyclic photophosphorylation does not produce ATP
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- 36.** What distinguishes photorespiration from cellular respiration? (1)
- A. Photorespiration produces ATP, whereas cellular respiration consumes it.
 - B. Photorespiration occurs in the light and does not produce ATP, unlike cellular respiration.
 - C. Photorespiration consumes CO₂ and releases O₂, the opposite of cellular respiration.
 - D. Photorespiration occurs in the mitochondria, while cellular respiration occurs in the chloroplasts.
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- 37.** What is the main function of the Electron Transport Chain (ETC) in cellular respiration? (1)
- A. To break down glucose into pyruvate and synthesize ATP
 - B. To facilitate the controlled release of energy from reduced cofactors.
 - C. To produce CO₂ through decarboxylation.
 - D. To synthesize glucose from CO₂ and H₂O
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- 38.** What happens during the reduction phase of the Calvin cycle? (1)
- A. CO₂ is combined with RuBP.
 - B. RuBP is regenerated using ATP.
 - C. 3-PGA is converted into G3P using ATP and NADPH.
 - D. Oxygen is released as a byproduct of water splitting
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- 39.** Why is the waxy cuticle important for a leaf's function? (1)
- A. It captures light energy for the mesophyll cells of plants.
 - B. It prevents the leaf from drying out and protects against infection.
 - C. It allows for the exchange of CO₂ and O₂.
 - D. It houses the chlorophyll pigments that absorb light.
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- 40.** What is the primary role of accessory pigments like carotenoids in photosynthesis? (1)
- A. To fix carbon dioxide into glucose.
 - B. To reflect all wavelengths of light to protect the leaf.
 - C. To absorb light energy and pass it to chlorophyll.
 - D. To absorb light energy and CO₂ for starting reaction
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- 41.** If a plant is kept in the dark for 24 hours, which process will stop first? (1)
- | | |
|---------------------|----------------------------------|
| A. The Calvin Cycle | B. The Light-Dependent Reactions |
| C. Glycolysis | D. Lactic Acid Fermentation |
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- 42.** If a metabolic toxin specifically binds to and inhibits the ATP Synthase in a cell, which of the following results would be observed? (1)
- A. The Electron transport chain would stop pumping protons immediately
 - B. The cell would be unable to produce ATP via oxidative phosphorylation
 - C. The cell would be unable to produce ATP via substrate level phosphorylation
 - D. The pH of the mitochondrial matrix would decrease significantly

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- 43.** Which plant would most likely thrive in a hot, dry desert by opening its stomata only at night? (1)
- A. Wheat
B. Maize
C. Rice
D. Pineapple
-
- 44.** If you were to measure the pH inside the thylakoid lumen during active photosynthesis, you would expect it to be: (1)
- A. Highly acidic (low pH) due to the accumulation of protons.
B. Highly alkaline (high pH) due to loss of protons.
C. Neutral (pH 7.0) due to equal proton concentration
D. Exactly the same as the stroma
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- 45.** What is the metabolic consequence for the citric acid cycle if the supply of oxaloacetate is depleted within the mitochondrial matrix? (1)
- A. The cycle will continue by using pyruvate as the primary 4-carbon acceptor
B. The cycle will cease because oxaloacetate is required to combine with Acetyl-CoA to form citric acid
C. The cycle will speed up to produce more lactic acid from pyruvate
D. The cycle will bypass the first step and directly produce FADH₂ from Acetyl-CoA.
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- 46.** How does the C₄ pathway reduce photorespiration compared to the C₃ pathway? (1)
- A. It uses Rubisco to fix carbon at night
B. It completely eliminates the need for the Calvin cycle.
C. It produces oxygen inside the bundle sheath cells than mesophyll cells.
D. It physically separates carbon fixation from the Calvin cycle using different cell types.
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- 47.** Which of the following is the correct movement of electrons during the light reaction of photosynthesis? (1)
- A. NADP⁺ → PS-II → PS-I → H₂O
B. H₂O → PS-I → PS-II → NADP⁺
C. H₂O → PS-II → PS-I → NADP⁺
D. H₂O → NADP⁺ → PS-I → PS-II
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- 48.** In the ETC, which complex is responsible for the final reduction of Oxygen to Water? (1)
- A. Complex I
B. Complex II
C. Complex III
D. Complex IV
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- 49.** Compare Alcoholic Fermentation and Lactic Acid Fermentation. What do they have in common? (1)
- A. Both produce CO₂.
B. Both regenerate oxidized NAD⁺ (NAD⁺)
C. Both produce 36 ATP per glucose.
D. Both occur in the mitochondria.
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- 50.** In an experiment, if O₂ levels are depleted in a eukaryotic cell, what is the most likely outcome for the Krebs Cycle? (1)
- A. It will speed up to compensate for lost energy.
B. It will stop because NAD⁺ and FAD cannot be regenerated by the ETC.
C. It will continue normally using CO₂ instead of O₂.
D. It will switch to producing glucose instead of breaking it down.
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- 51.** Why is the inner mitochondrial membrane highly folded into 'cristae'? (1)
- A. To decrease the volume of the matrix.
B. To protect the DNA from oxygen.
C. To make the mitochondria look more complex.
D. To increase surface area for ETC and ATP production.

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- 52.** What is the evidence that supports that Photorespiration is a wasteful process? (1)
- A. It prevents the plant from absorbing light.
 - B. It only occurs in dead plants.
 - C. It consumes ATP and releases CO₂ without producing sugar
 - D. It creates too much amino acid for the plant to store.
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- 53.** If you were to engineer a C₃ crop (like wheat) to be more drought-resistant, which modification would be most effective? (1)
- A. Removing all stomata from the leaves to reduce transpiration
 - B. Decreasing the rate of photorespiration to save water
 - C. Making the leaves larger to catch more rain.
 - D. Introducing the PEPCO enzyme and bundle sheath anatomy of C₄ plants
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- 54.** How would you modify the Z-scheme of photosynthesis if Photosystem II was disabled by a toxin? (1)
- A. The plant would only perform cyclic photophosphorylation using PS I.
 - B. The plant would produce more CO₂ without producing any ATP.
 - C. The plant would start using CO₂ to split water.
 - D. The Calvin cycle would work twice as fast.
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- 55.** Which of the following is an example of a homologous structure? (1)
- | | |
|---|--|
| A. Human arm and whale flipper | B. Wing of a bird and wing of a mosquito |
| C. Eye of an octopus and eye of a human | D. Tail of a fish and tail of a scorpion |
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- 56.** Which theory of the origin of life suggests that life arrived on Earth from outer space in the form of spores? (1)
- | | |
|----------------------------------|---------------------------|
| A. Special Creation | B. Spontaneous Generation |
| C. Cosmozoan Theory (Panspermia) | D. Biochemical Theory |
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- 57.** Natural Selection leads to evolution: (1)
- A. By creating new mutations on demand.
 - B. By ensuring that all individuals in a population survive and reproduce more successfully
 - C. By allowing individuals with favorable traits to survive and reproduce more successfully.
 - D. By changing the environment to fit the organisms to reproduce more successfully
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- 58.** Scientists use radiometric dating to determine the age of fossils. If a fossil contains a radioactive isotope with a half-life of 10,000 years, and only 25% of the original isotope remains, what is the approximate age of the fossil? (1)
- | | |
|-----------------|-----------------|
| A. 10,000 years | B. 20,000 years |
| C. 30,000 years | D. 40,000 years |
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- 59.** In Redi's experiment, why did maggots not appear in the jar covered with cloth? (1)
- | | |
|--------------------------|---|
| A. The meat was poisoned | B. Flies could not lay eggs on the meat |
| C. The meat did not rot | D. Air was completely excluded |

— END OF EXAMINATION —